

Validation Report

Default Campaign

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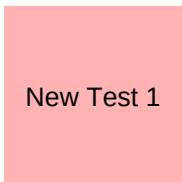
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Preamble

Default description. Please edit the campaign description.

Result Overview



Summary of Tests in Suite "New Suite 0"

1 Suite: New Suite 0

Suite Status: Failed

Default description. Please edit the suite description.

1.1 Test: New Test 1

Test Status: Failed



Default description. Please edit the test description.

1.1.1 Sample: CAVM Min/Max

Acceptance Status: Passed

Loaded Data:

- Point Cloud Data (PCD) 1

File	SH256 Hash
Real_van.pcd	6f9576cca4eeff638e439b5aacd1139ca9b7b3e29f5d3ea9c23f902924106db0

- Point Cloud Data (PCD) 2

File	SH256 Hash
Sim_Van.pcd	2b71e6e26ced59bc1f15073d975c2f918610df6f8f382b6f3b4e438e3064a99e

Applied Metric: CAVM

Inputs:

- Point Cloud Data (PCD) 1
- Point Cloud Data (PCD) 2

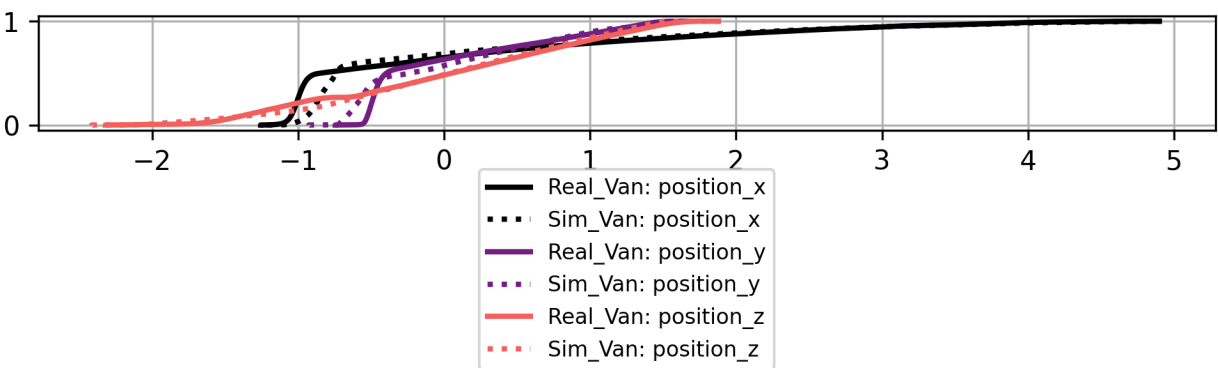


Figure 1.1.1-1: Metric 'CAVM': eCDF for Real_Van vs Sim_Van (frame 0).

Applied Acceptance: Min/Max

Frame Real_Van	Frame Sim_Van	Metric Result Name	Value	Threshold [Min, max]
Real_van	Sim_Van	position_x position_y position_z	0.1503 0.0985 0.0836	[0.0000, 0.2000]

Table 1.1.1-1: Comparison of the metric value with the defined thresholds.

1.1.2 Sample: Bias Min/Max

Acceptance Status: **Failed**

Loaded Data:

- Point Cloud Data (PCD) 1

File	SH256 Hash
Real_van.pcd	6f9576cca4eeff638e439b5aacd1139ca 9b7b3e29f5d3ea9c23f902924106db0

- Point Cloud Data (PCD) 2

File	SH256 Hash
Sim_Van.pcd	2b71e6e26ced59bc1f15073d975c2f918 610df6f8f382b6f3b4e438e3064a99e

Applied Metric: Bias

Inputs:

- Point Cloud Data (PCD) 1
- Point Cloud Data (PCD) 2

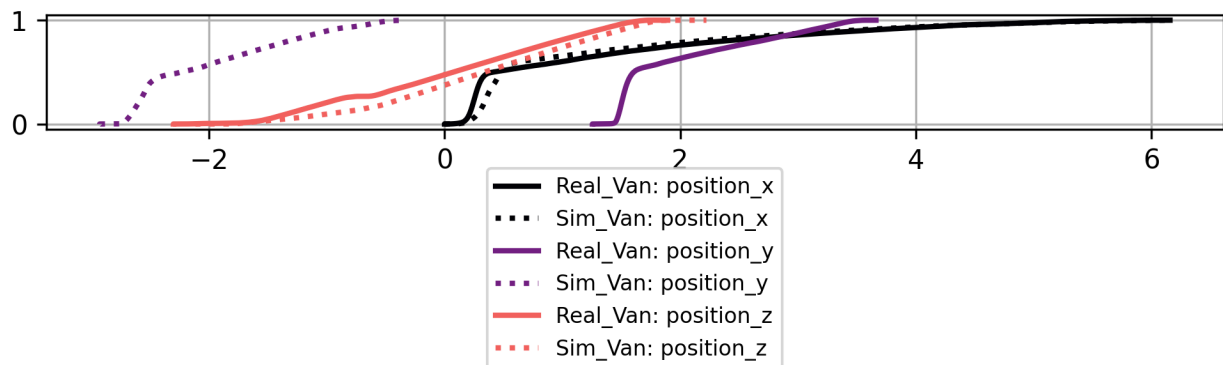


Figure 1.1.2-1: Metric 'Bias': eCDF for Real_Van vs Sim_Van (frame 0).

Applied Acceptance: Min/Max

Frame Real_Van	Frame Sim_Van	Metric Result Name	Value	Threshold [Min, max]
Real_van	Sim_Van	position_x position_y position_z	-0.0449 -4.0000 0.2789	[-0.0500, 0.0500]

Table 1.1.2-1: Comparison of the metric value with the defined thresholds.

Appendix A: Data Loader Explanation

PCD

This data loader module loads .pcd files containing 3D perception data, following the Point Cloud Data (PCD) format used by the Point Cloud Library (PCL). When multiple files are provided, they are combined as sequential frames. All files must have the same field structure and version to ensure compatibility. Only the fields x, y, z, intensity (lidar), intensity (lidar) and rcs (radar) are supported (case-insensitive).

Appendix B: Metric Explanation

CAVM

This metric module calculates deviation between two distribution functions via the Area Validation Metric, which are corrected by their mean value. CAVM is short for corrected Area Validation Metric. This metric module calculates the deviation in the spread of two distribution functions. As a first step, the Bias is calculated for the data of the two inputs. Afterward, the eCDF of input 2 is corrected by the Bias. Then, the Area Validation Metric (AVM) is calculated by calculating the area in between the ECDF of input 1 and the corrected ECDF of input 2.

Bias

This metric module calculates the estimate of the mean value deviation between the two channels based on distribution functions. As a first step, an empirical cumulative distribution function (eCDF) is calculated for each input. Then, the Area Validation Metric (AVM) is calculated by calculating the signed area in between the ECDF of the two inputs. Thereby, the area above the ECDF of the second input is subtracted from the area below the ECDF of the second input.

Appendix C: Acceptance Explanation

MinMax

This acceptance module applies minimum and maximum thresholds to the results of the metric within every time frame. A threshold is defined as minimum and maximum values.